Lecture 15: Maps

Heather Mattie, PhD

October 21, 2025

Recipe of the Day!

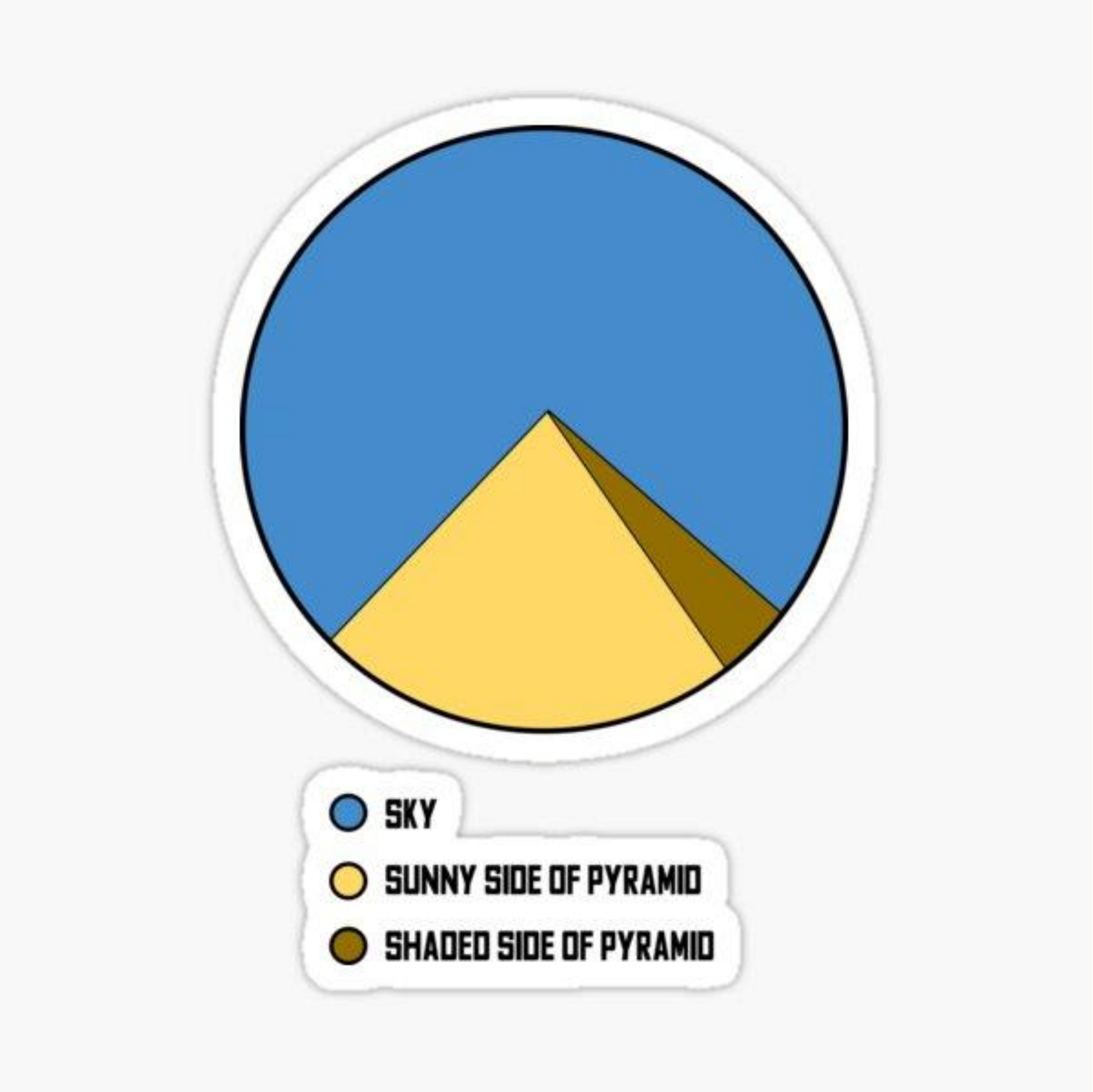
[Diwali Dinner Recipes]

[Connors Farm]

Danvers, MA

Announcements

- Homework #3 due Friday, October 24 by 11:59pm
- No lab this week
- Midterm will be released on **Monday, October 27** and due **Friday, November 7**
  - No late days may be used
  - Generative AI **may not** be used **in any way**



Homework reminders

- If something is unclear, please ask in office hours, on canvas, or via email
- Generative AI may be used to help with debugging or explaining a concept, but you should not copy and paste code from it and need to cite when and how you used it

Mid-course survey results

- 45% response rate
- Useful for learning
  - In-class coding
  - Lectures and lab are recorded
  - Lecture slides, RMarkdown notes
  - Pacing
  - Learning environment

Mid-course survey results

- Areas for improvement
  - Too much time spent on debugging student code in class
    - Too much time spent setting up GitHub in class
  - More coding examples and/or cheat sheets
  - Mentioning which lectures should be referenced for which homework assignment
  - Rubric for homework assignments
  - More guidance on how to use Rmd files in course repository
  - Less time for slides, more time for in-class coding

Coding question of the day

Using the **`gapminder`** dataset, make a line graph of life expectancy over time.

Include one line per country, and another line with the global average life expectancy.

Be sure to label the global average line and to make it stick out among the country-specific lines. Use the code below to calculate the crude global average for each year.

Hints:

1. Use alpha to help make the global average line stick out.
2. Use 2 **`geom_line`** layers.
3. Note that **`avg`** is a data frame.

Be sure to run this code first

```
library(dslabs)
library(dplyr)
library(ggplot2)
data("gapminder")

avg <- gapminder %>%
  group_by(year) %>%
  summarize(global_avg = mean(life_expectancy))
```

Maps

Maps Roadmap

1. The **`maps`** package
2. Plotting global life expectancy
3. Plotting US state murder rates
4. Case study: COVID-19 cases

Maps

The `maps` package

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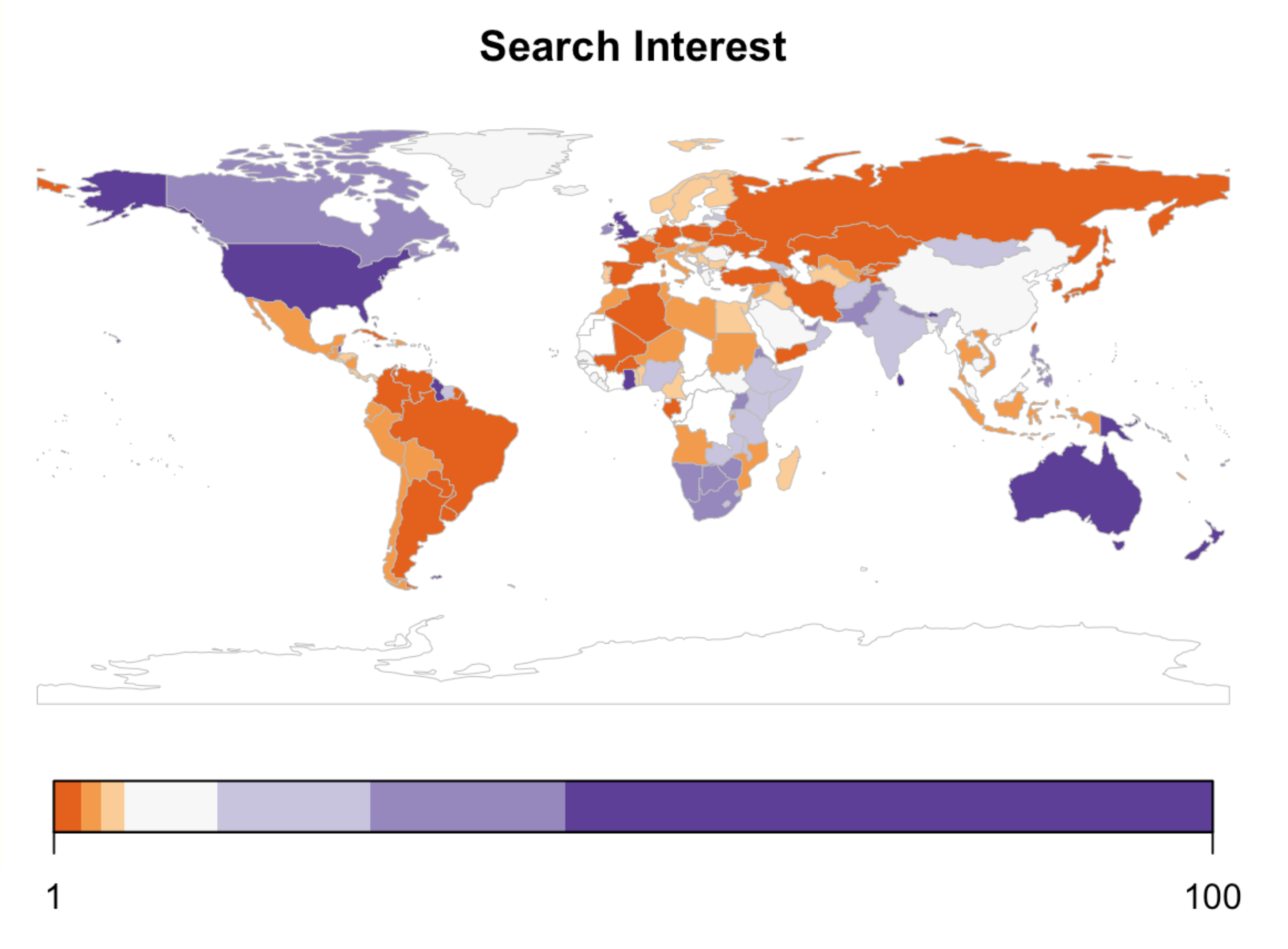
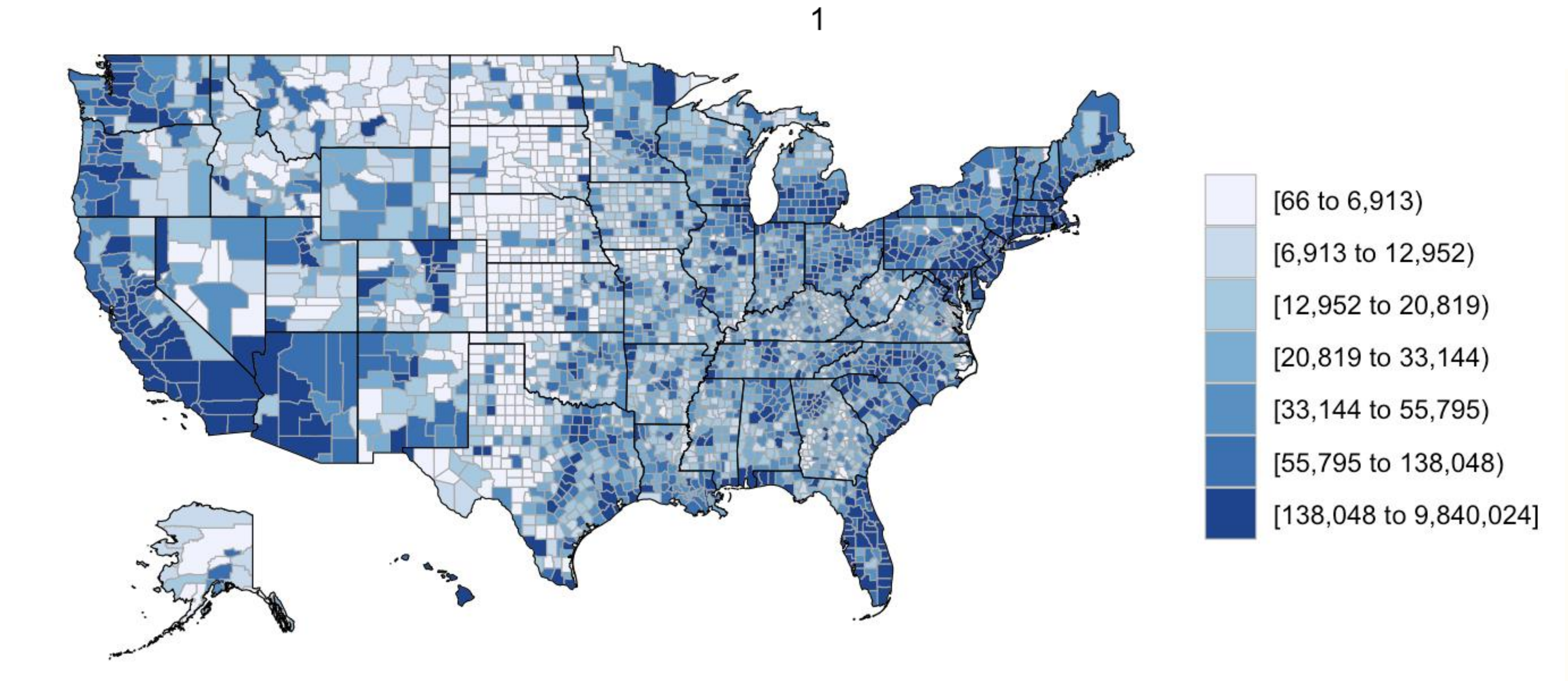
Maps

- Plots of maps can be very powerful, informative and aesthetically pleasing
- They can also be misleading if not created correctly
- We will us the **`maps`** package to create plots of maps
- Other packages: **`ggspatial`**, **`sf`**, **`urbnmapr`**, **`rnaturalearth`**, etc.

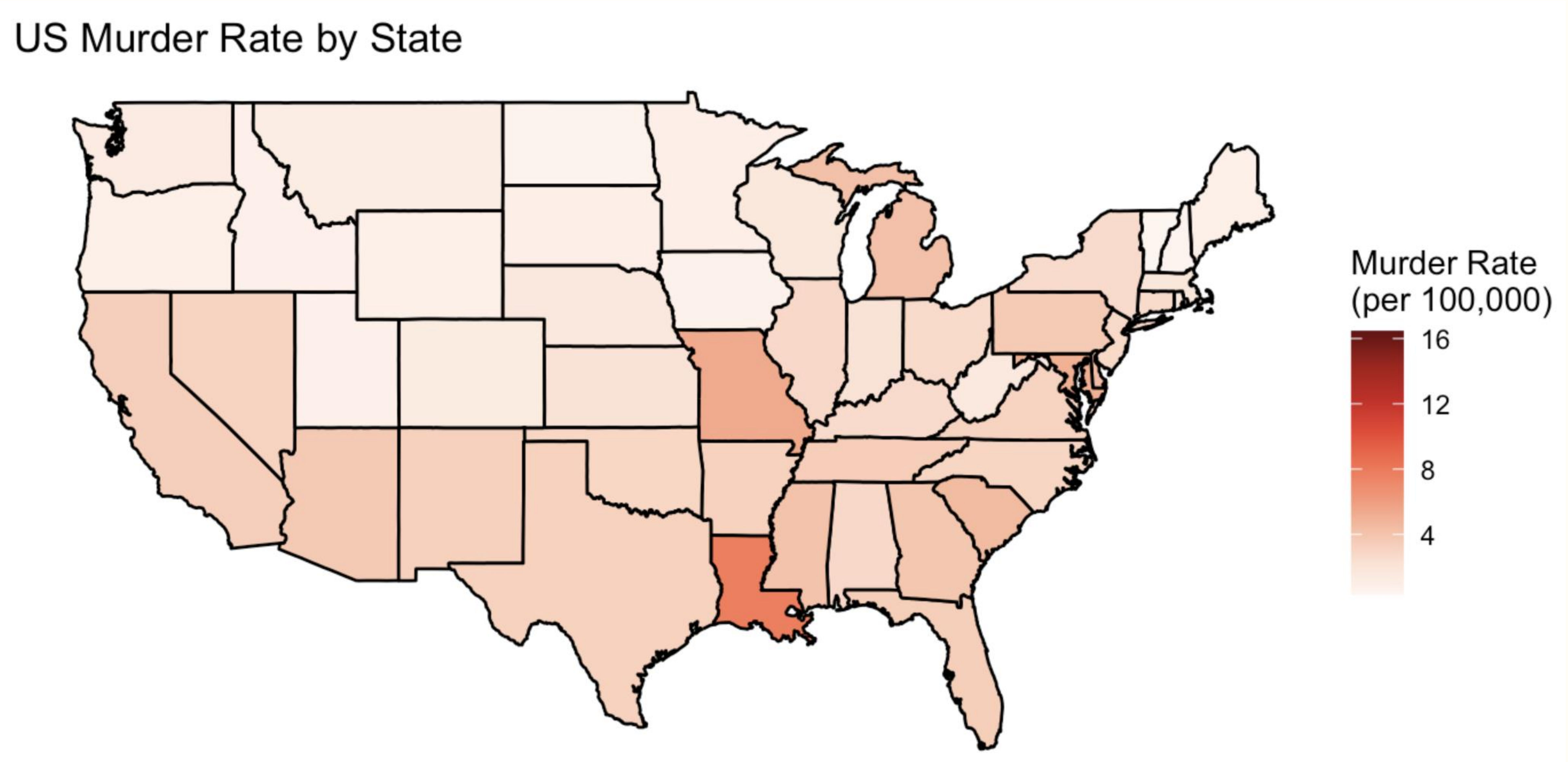




`maps`

The maps package contains outlines of continents, countries, states, and counties / jurisdictions / provinces

- Specify which map to use with `map_data()`
- Loaded as a data frame and can be used with the pipe (`%>%`) and `ggplot`



map_data("world")

- `long` = longitude
- `lat` = latitude
- `group` indicates which countries can be drawn together
- `order` indicates which order to connect the `lat` and `long` dots
- `region` = country
- `subregion` = county, political jurisdictions, etc.

```r
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##        long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba      <NA>
## 2 -69.89571 12.42300     1     2  Aruba      <NA>
## 3 -69.94219 12.43853     1     3  Aruba      <NA>
## 4 -70.00415 12.50049     1     4  Aruba      <NA>
## 5 -70.06612 12.54697     1     5  Aruba      <NA>
## 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

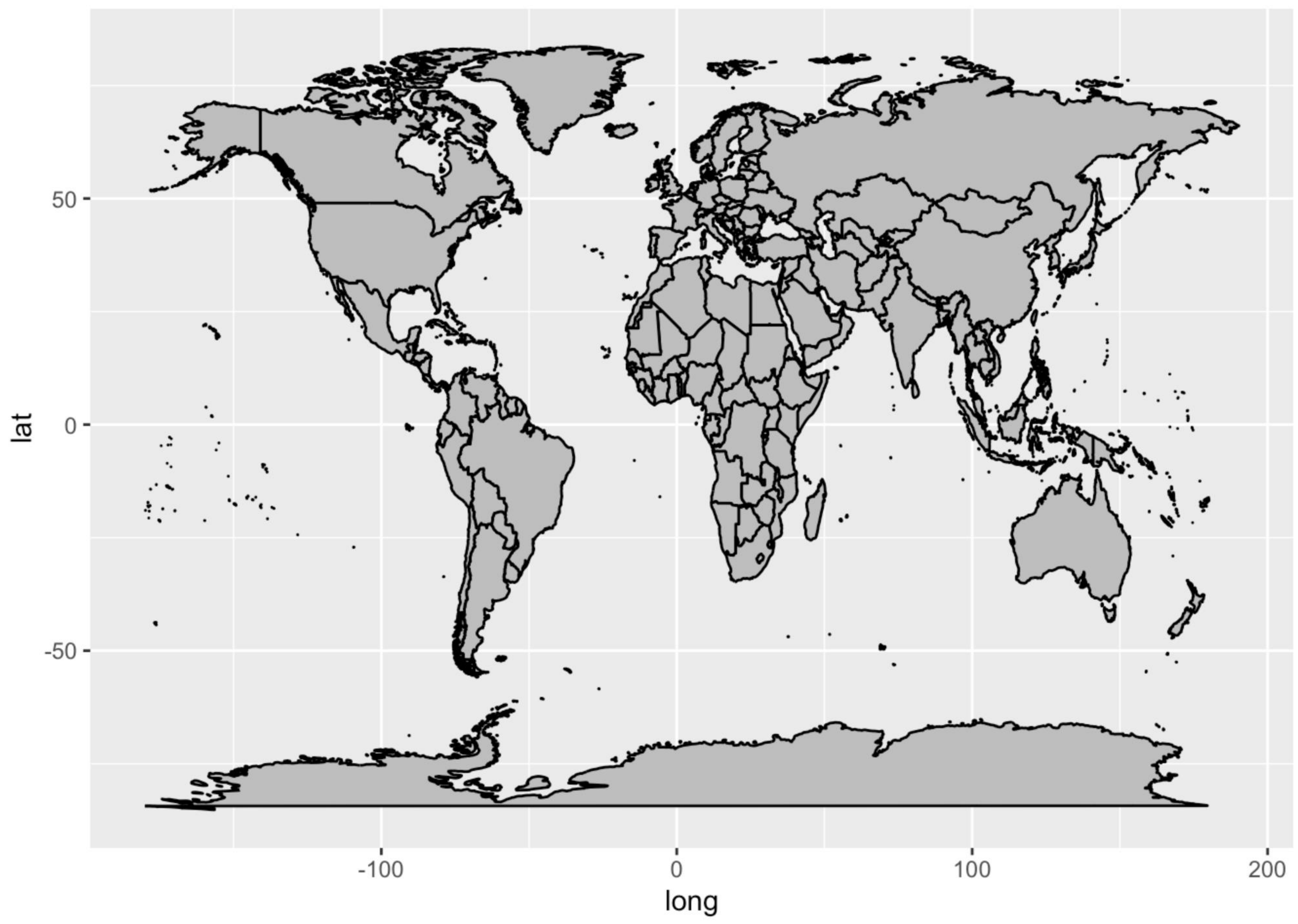
```r
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```

map_data(“world”)

```r
library(tidyverse)
library(maps)
world_map <- map_data("world")
head(world_map)
```

```
##        long      lat group order region subregion
## 1 -69.89912 12.45200     1     1  Aruba      <NA>
## 2 -69.89571 12.42300     1     2  Aruba      <NA>
## 3 -69.94219 12.43853     1     3  Aruba      <NA>
## 4 -70.00415 12.50049     1     4  Aruba      <NA>
## 5 -70.06612 12.54697     1     5  Aruba      <NA>
## 6 -70.05088 12.59707     1     6  Aruba      <NA>
```

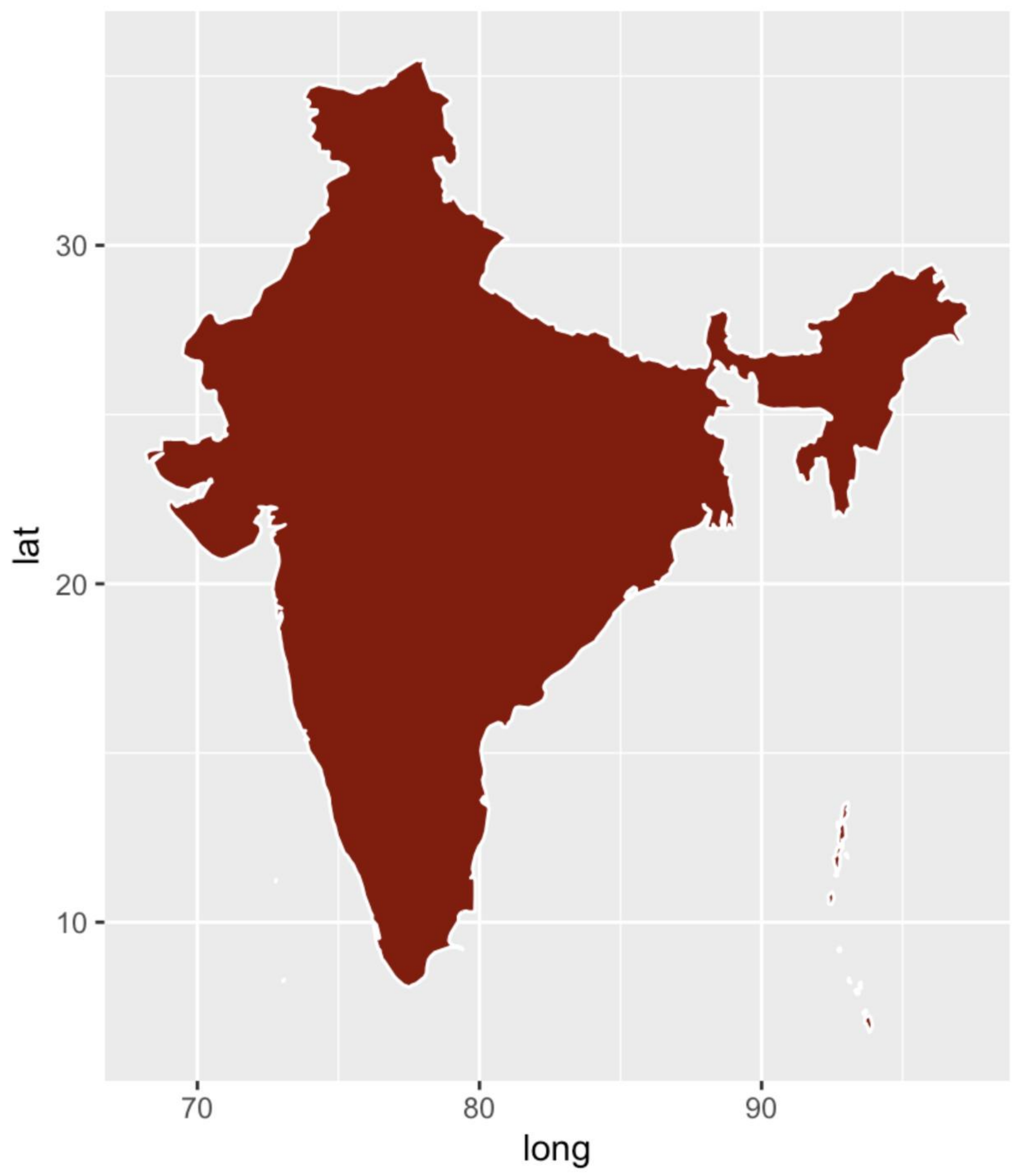
```r
world_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "gray", color = "black")
```



Countries

```
india_map <- map_data("world", "India")

india_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.2)
```

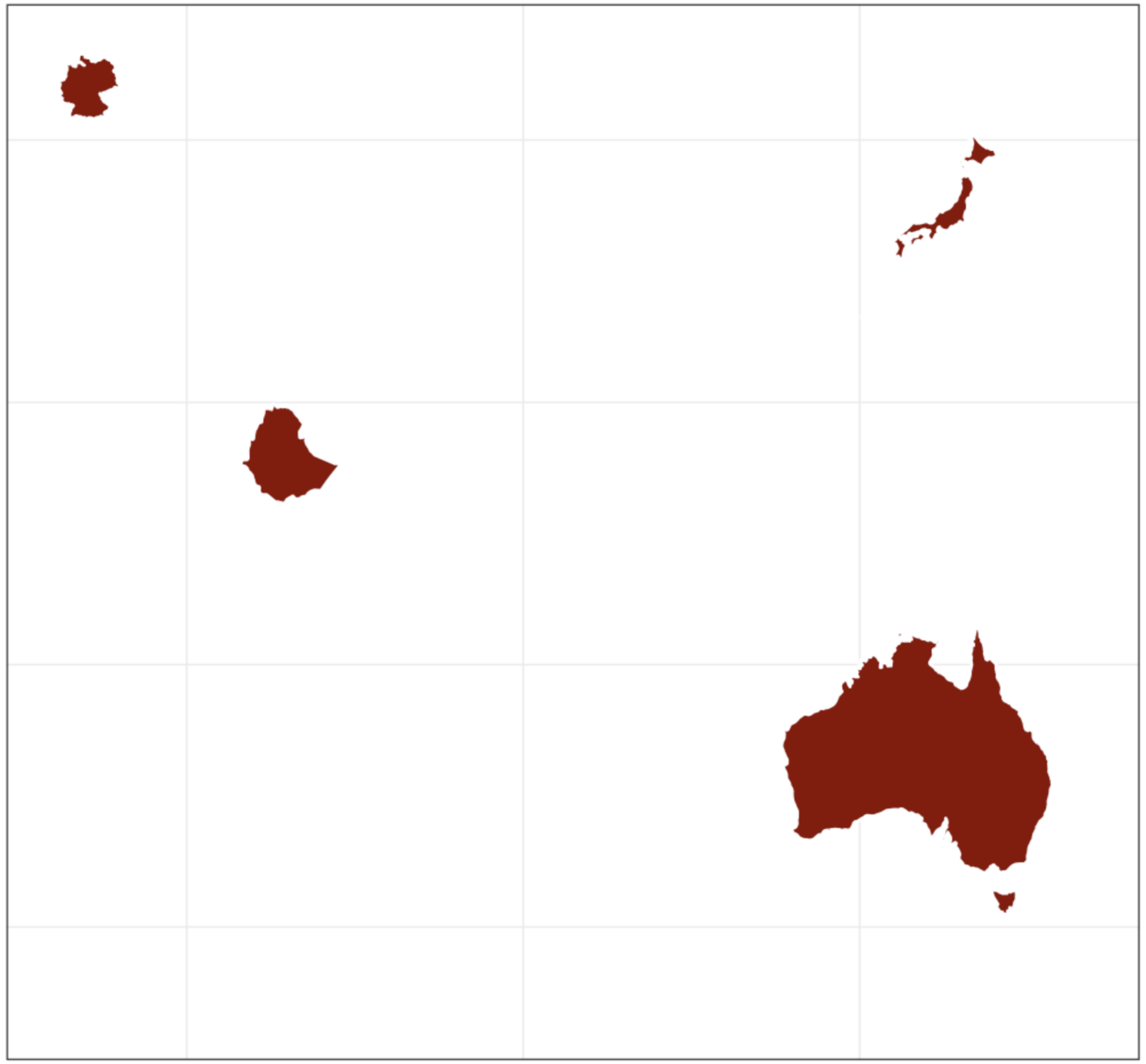


Countries

```r
theme_set(theme_bw())

multiple_countries_map <- map_data("world",
                                   c("Japan", "Ethiopia", "Australia", "Germany"))

multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  coord_fixed(1.3)
```



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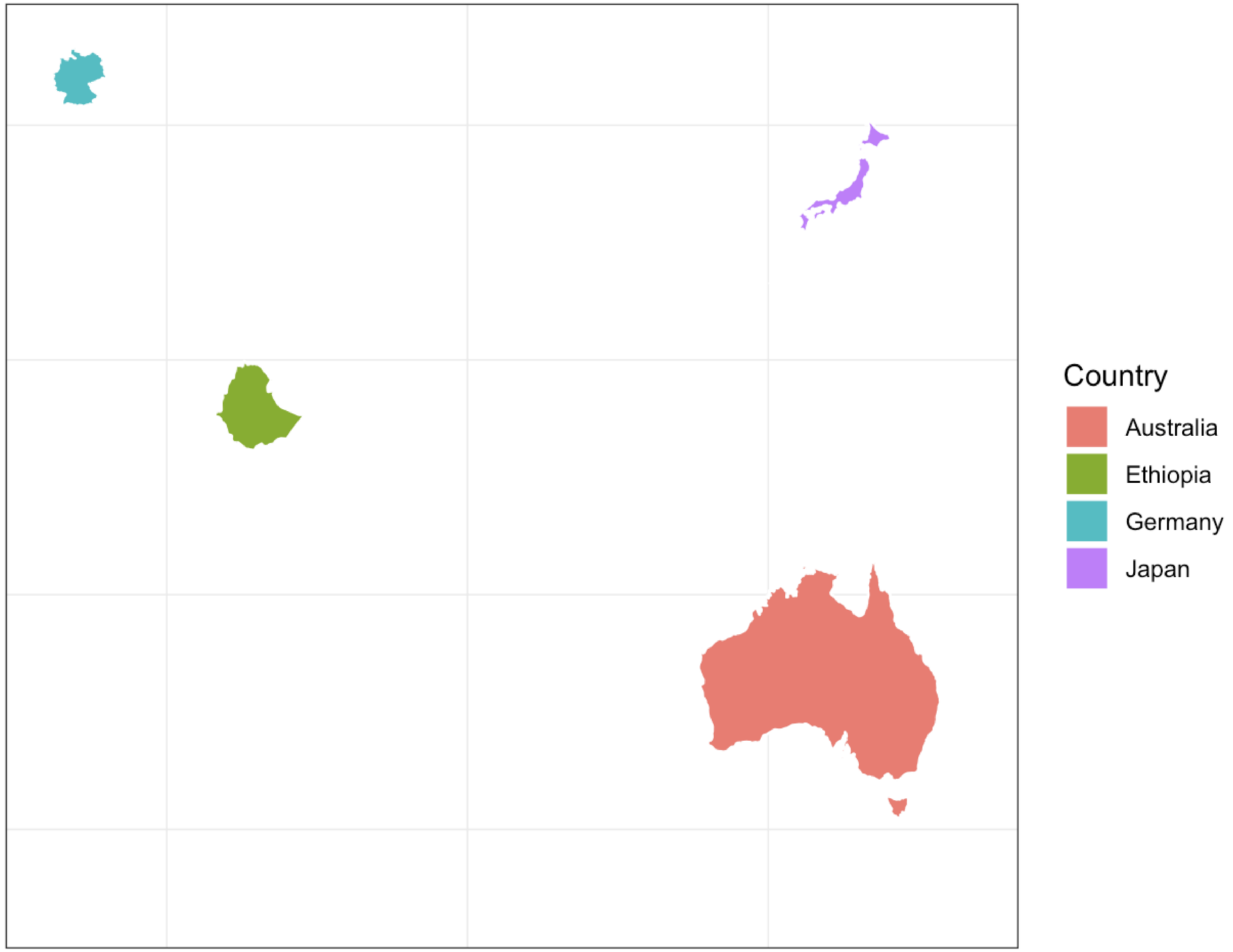
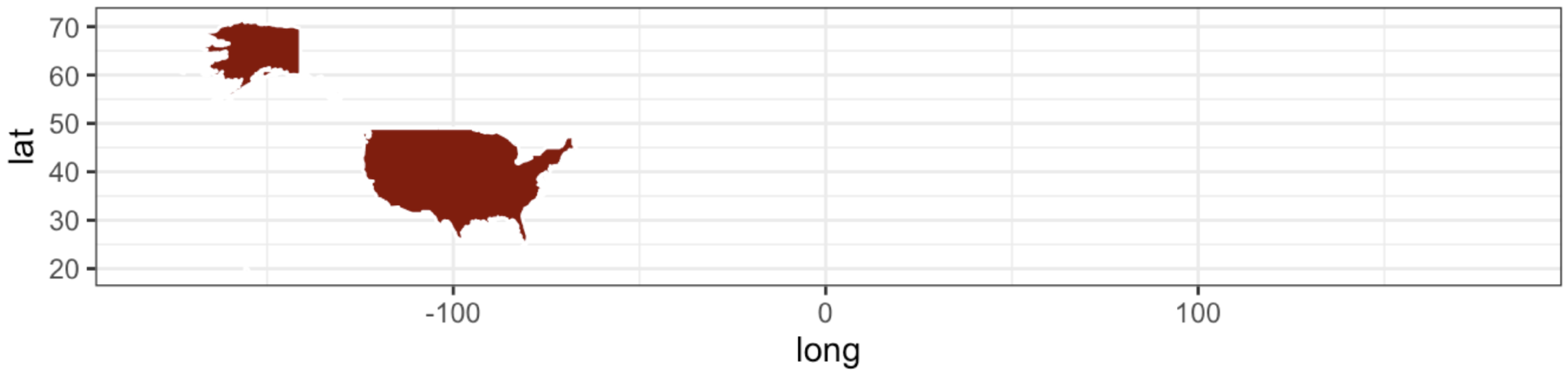
Countries

```
multiple_countries_map %>%
  ggplot(aes(x = long, y = lat, group = group, fill = region)) +
  geom_polygon(color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  labs(fill = "Country") +
  coord_fixed(1.3)
```



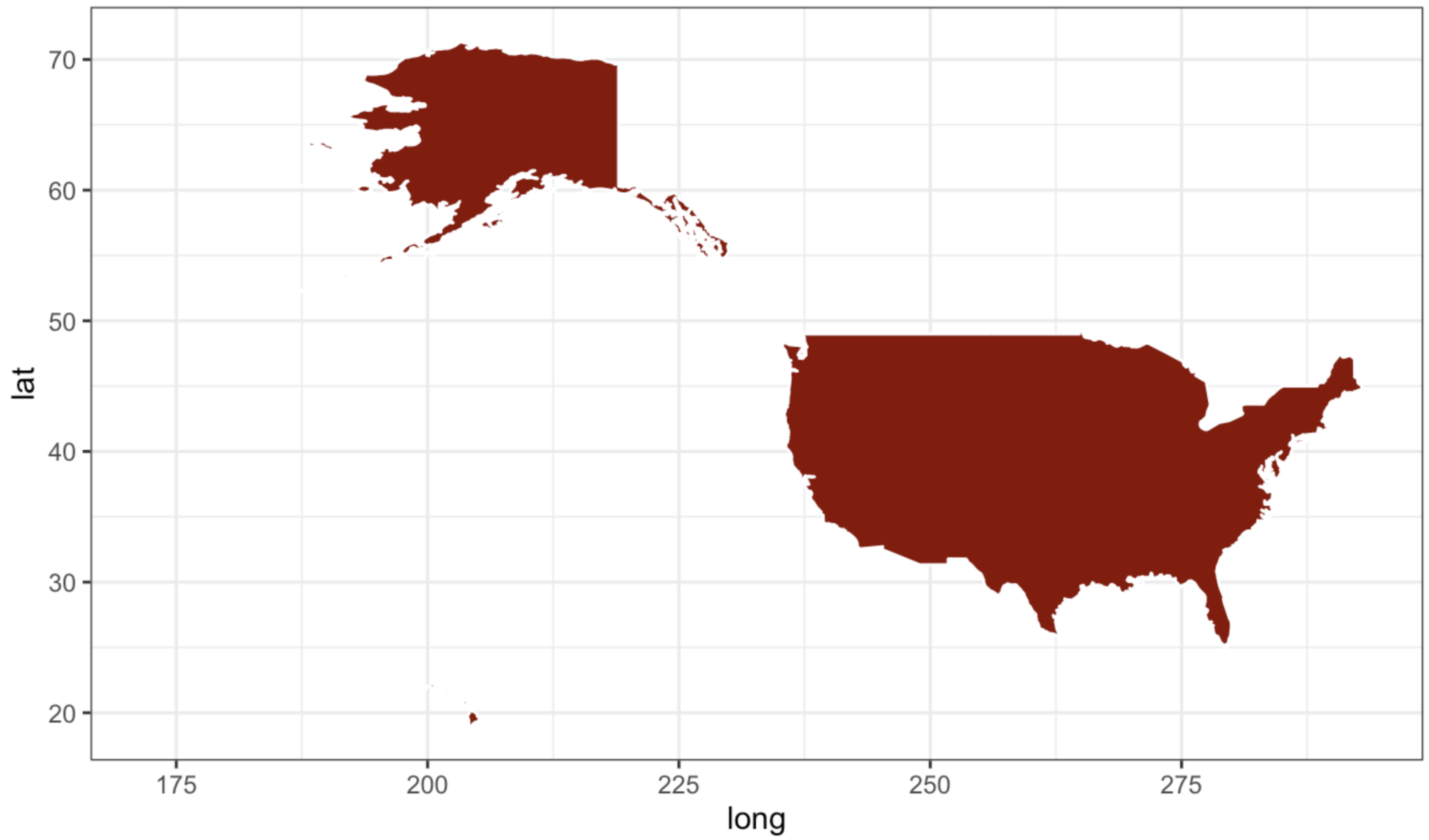
US maps

```
us_map <- map_data("world", "usa")

us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



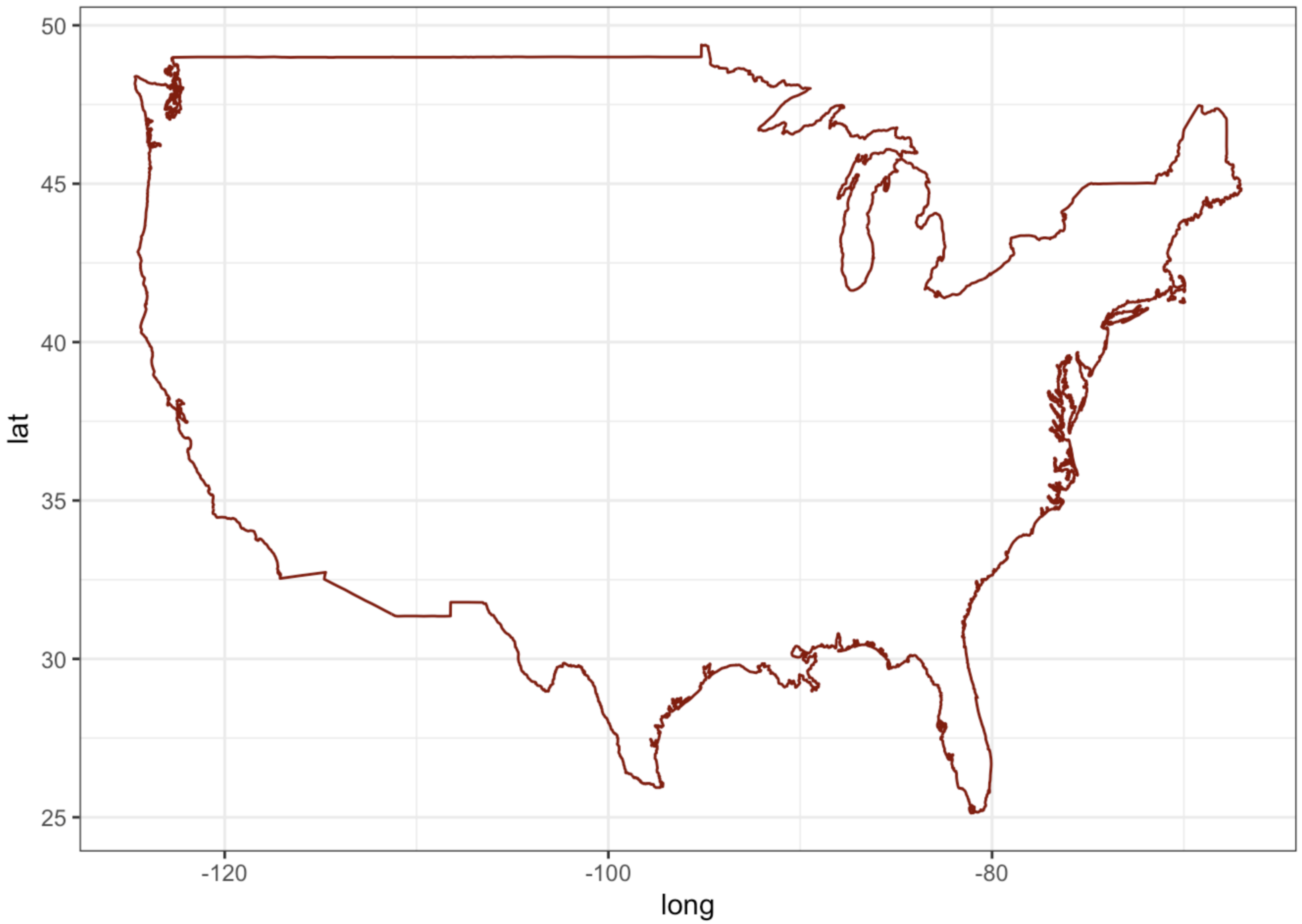
US maps

```
better_us_map <- map_data("world2", "USA")

better_us_map %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(fill = "darkred", color = "white") +
  coord_fixed(1.3)
```



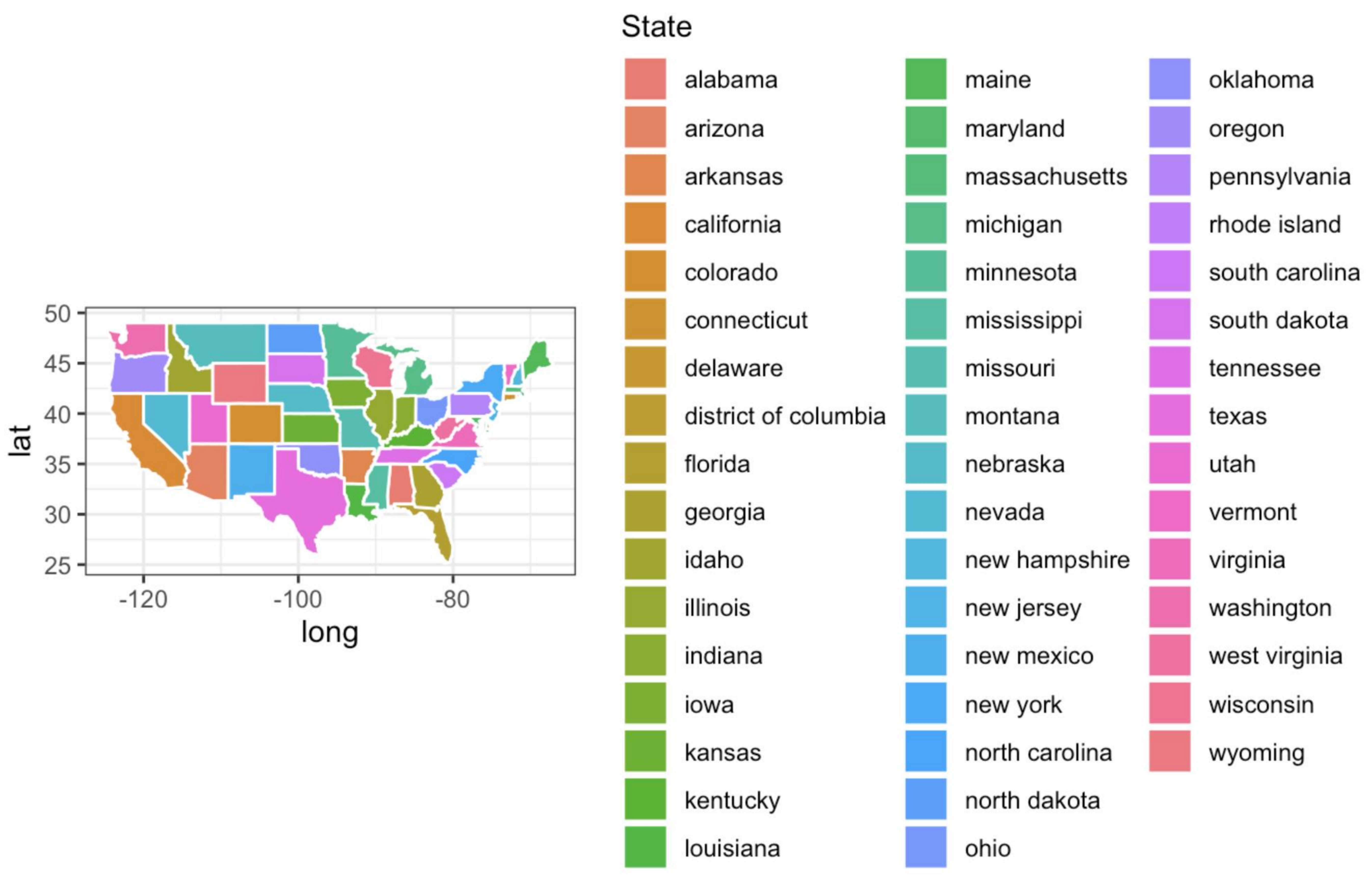
US maps

```
usa <- map_data("usa")

usa %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



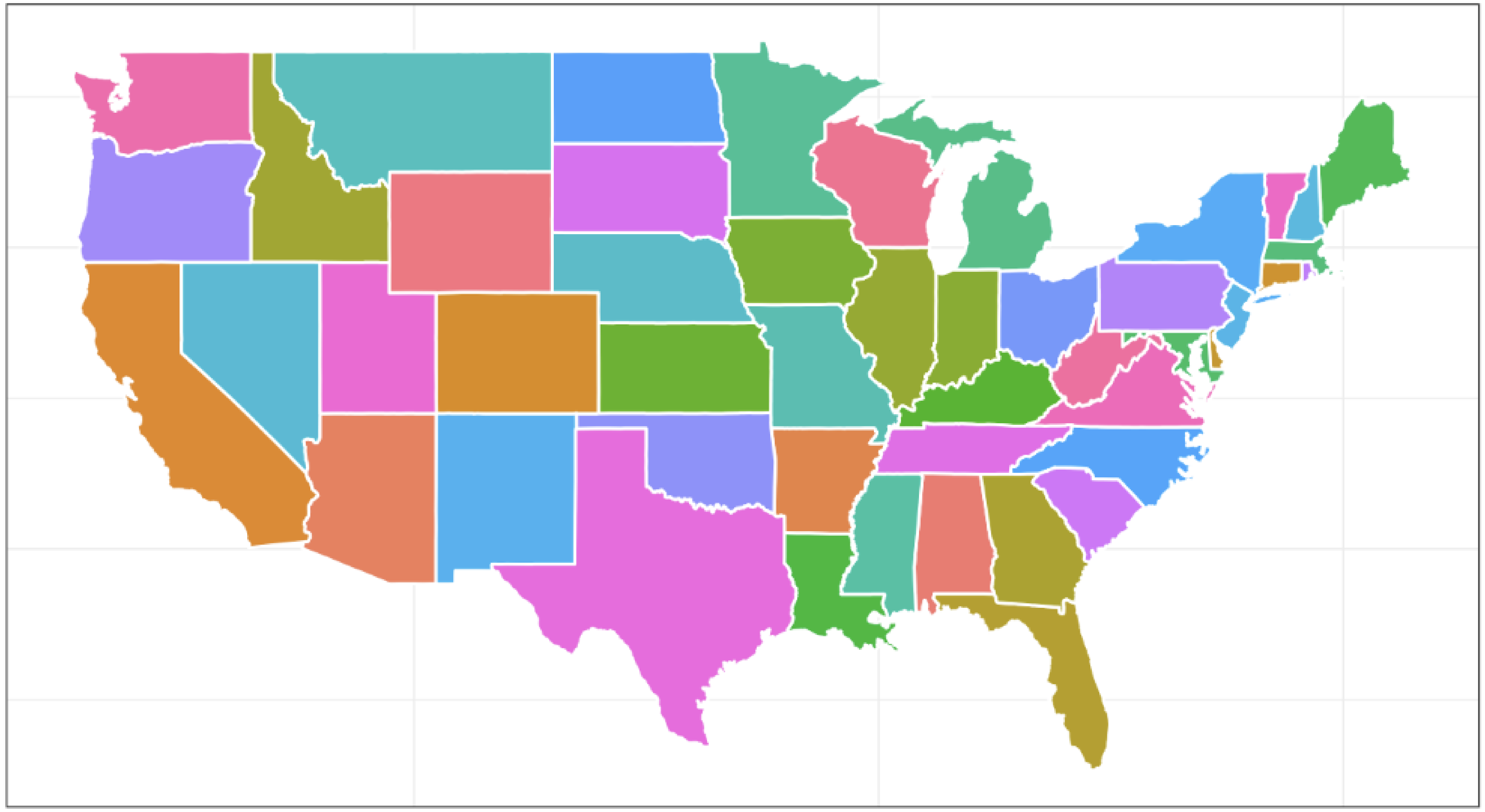
US states

```
us_states <- map_data("state")

us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  labs(fill = "State") +
  coord_fixed(1.3)
```



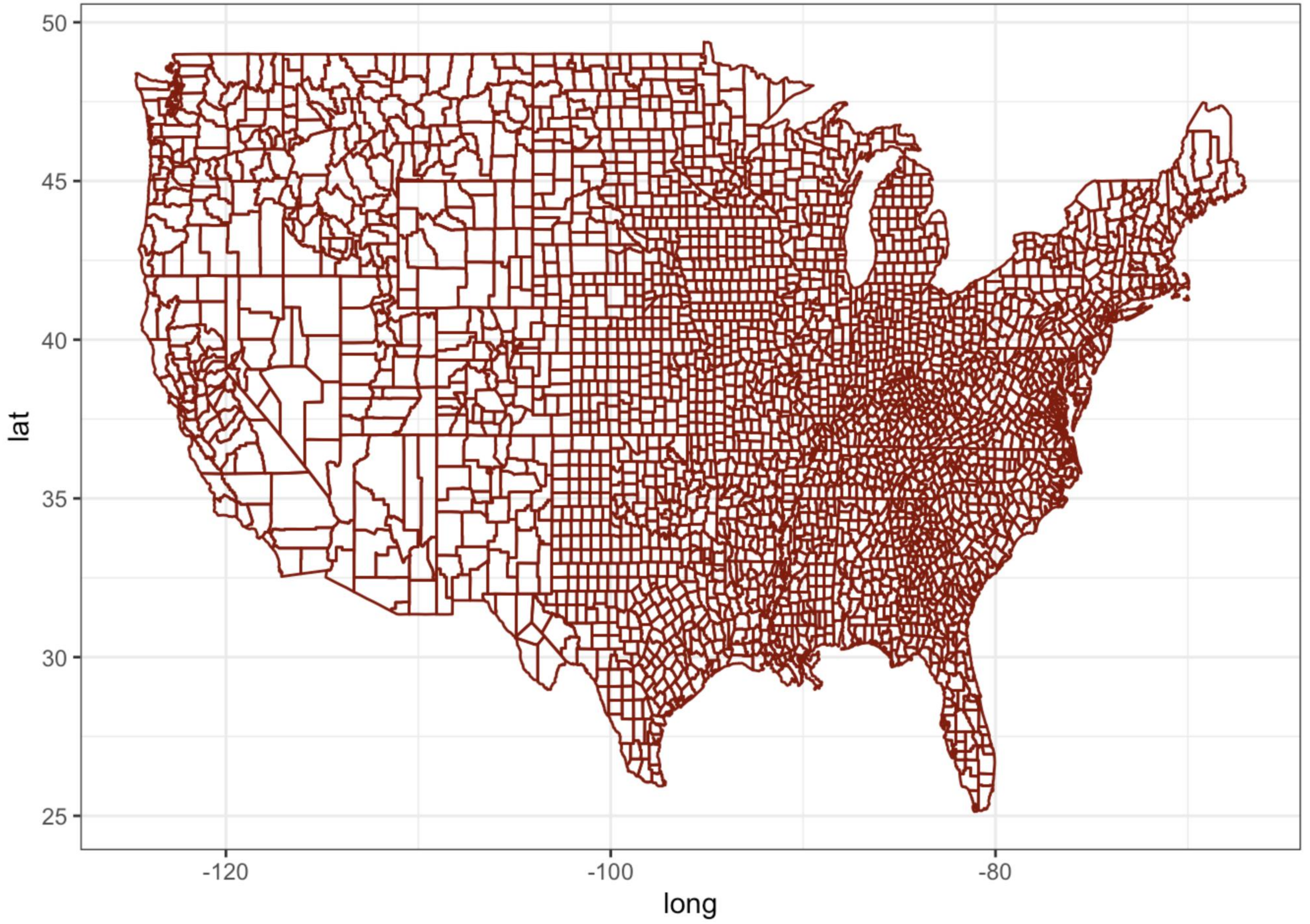
US states

```
us_states %>%
  ggplot(aes(x = long, y = lat, fill = region, group = group)) +
  geom_polygon(color = "white") +
  theme(panel.grid.major = element_blank(),
  panel.background = element_blank(),
  axis.title = element_blank(),
  axis.text = element_blank(),
  axis.ticks = element_blank()) +
  coord_fixed(1.3) +
  guides(fill = "none")
```



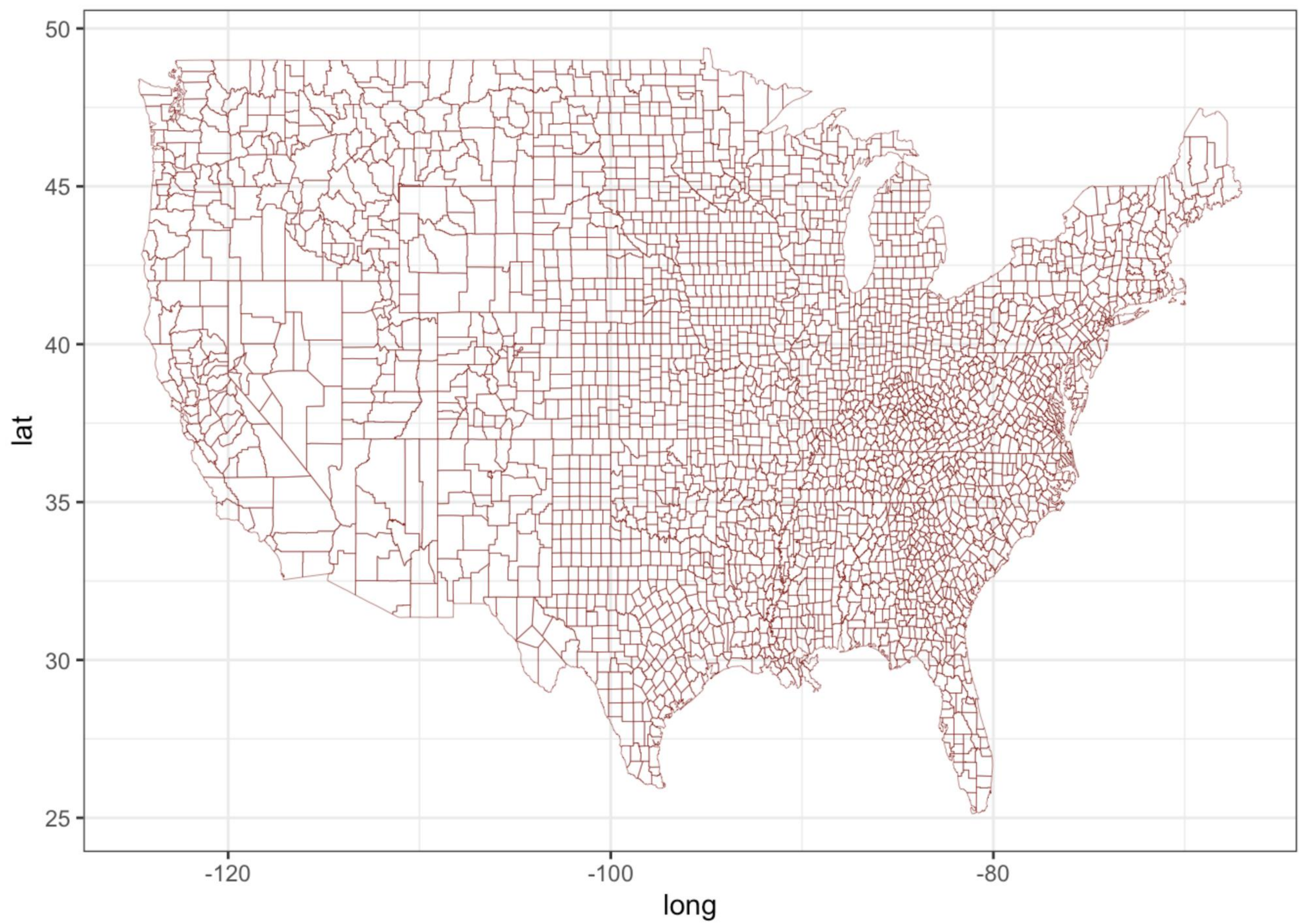
US counties

```r
AllCounty <- map_data("county")

AllCounty %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA)
```



US counties

```
AllCounty %>%
  ggplot(aes(x = long, y = lat, group = group)) +
  geom_polygon(color = "darkred", fill = NA, linewidth = .1)
```



Summary

- The `maps` package can be used to produce maps of continents, countries, states, counties, and provinces
- The `map_data()` function loads map data
- Color or labels should be used instead of a legend if there are too many categories (countries, states, etc.) represented in the map

Maps

Plotting global life expectancy

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Plotting global life expectancy

Task: plot a map displaying the life expectancy in 2015 for countries with data in the `gapminder` dataset

```r
library(plyr)
library(tidyverse)
library(maps)
library(dslabs)

data(gapminder)
gapminder_2015 <- gapminder %>%
  filter(year == 2015)

world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015,
                          by = c("region" = "country"))

head(merged_world)
```

```
##        long      lat group order region subregion year infant_mortality
## 1 -69.89912 12.45200     1     1  Aruba      <NA> 2015               NA
## 2 -69.89571 12.42300     1     2  Aruba      <NA> 2015               NA
## 3 -69.94219 12.43853     1     3  Aruba      <NA> 2015               NA
## 4 -70.00415 12.50049     1     4  Aruba      <NA> 2015               NA
## 5 -70.06612 12.54697     1     5  Aruba      <NA> 2015               NA
## 6 -70.05088 12.59707     1     6  Aruba      <NA> 2015               NA
##   life_expectancy fertility population gdp continent  region.y
## 1           75.72      1.66     103889  NA  Americas Caribbean
## 2           75.72      1.66     103889  NA  Americas Caribbean
## 3           75.72      1.66     103889  NA  Americas Caribbean
## 4           75.72      1.66     103889  NA  Americas Caribbean
## 5           75.72      1.66     103889  NA  Americas Caribbean
## 6           75.72      1.66     103889  NA  Americas Caribbean
```


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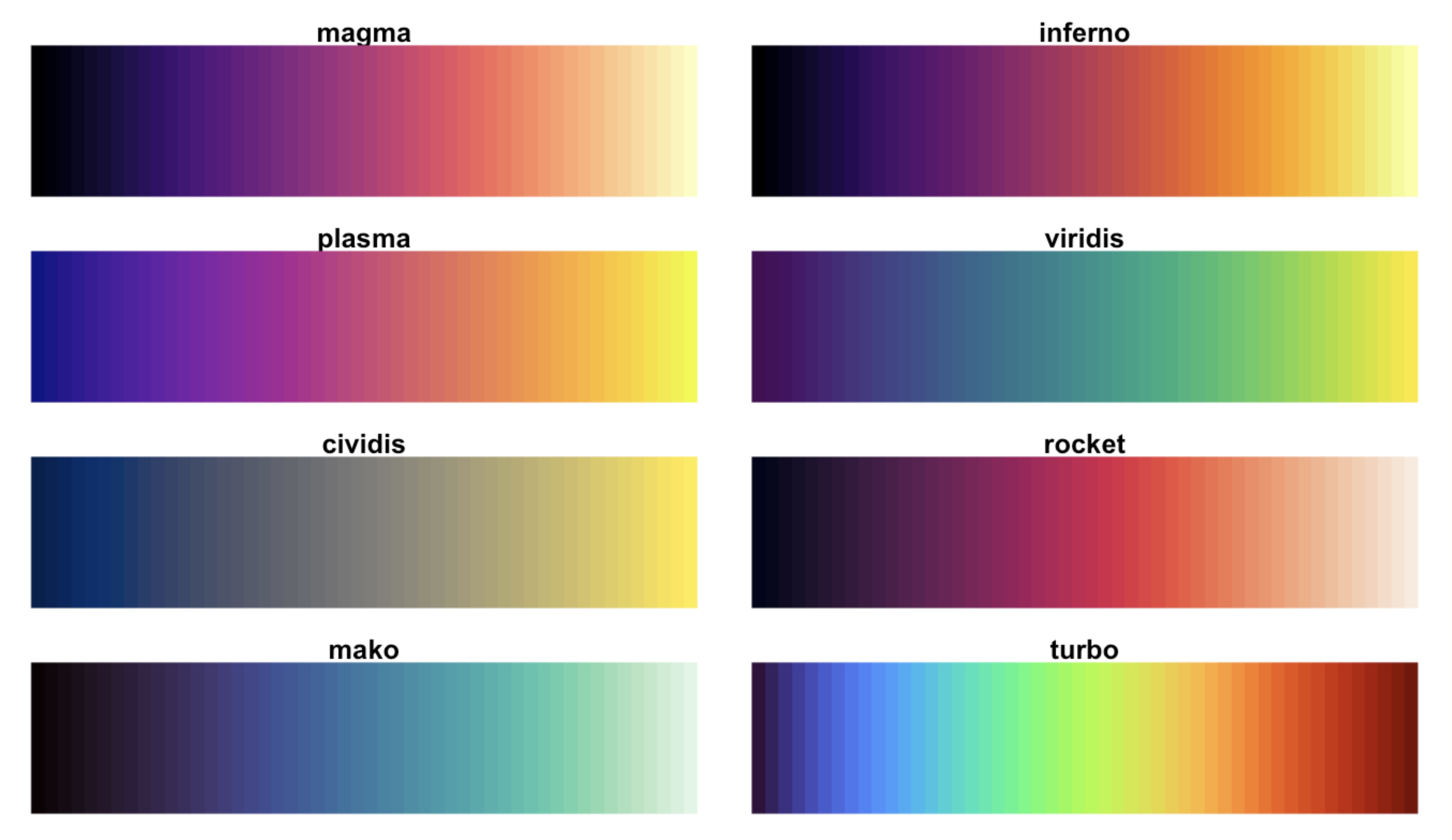
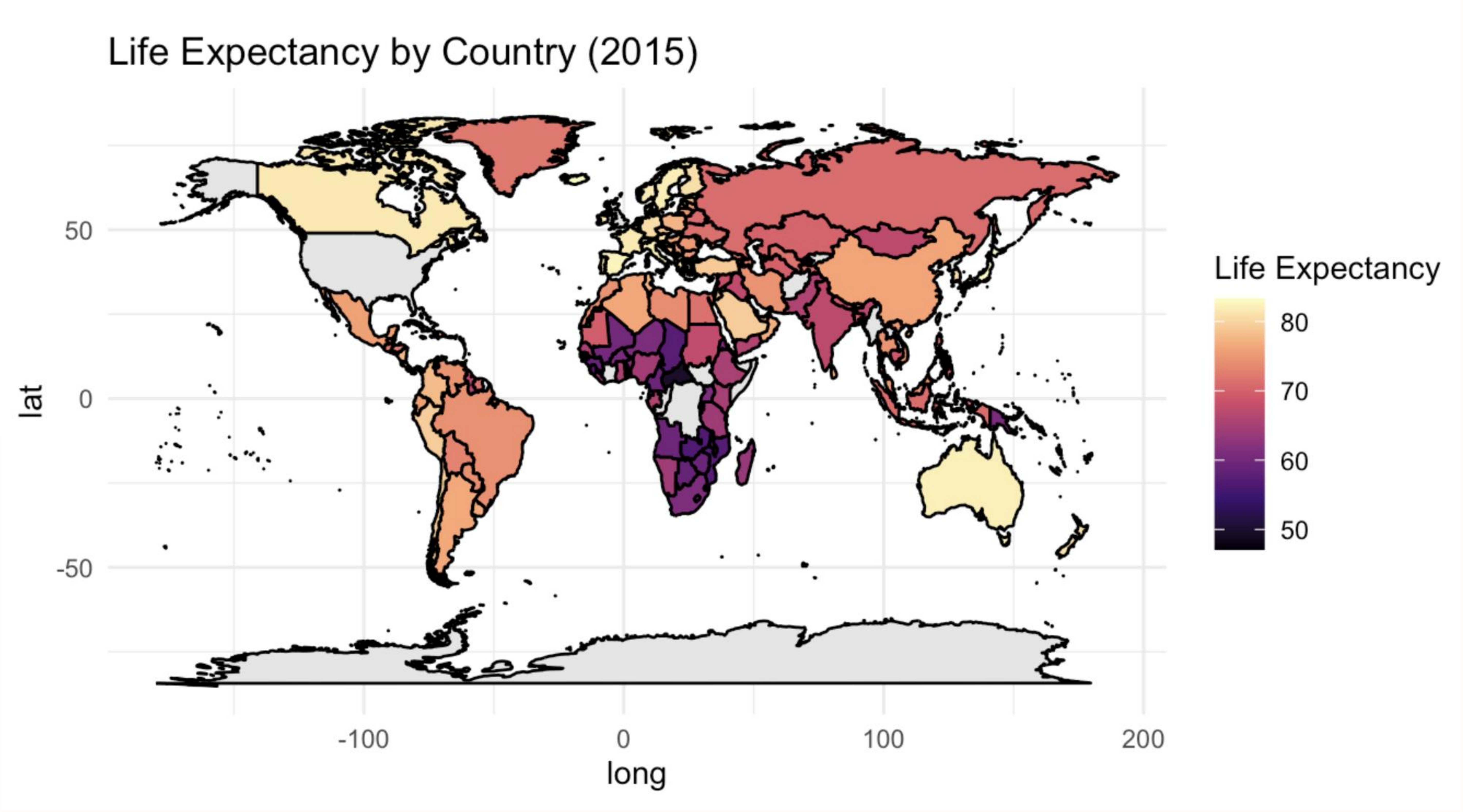
Plotting global life expectancy

```
ggplot(merged_world, aes(long, lat, group = group,
                         fill = life_expectancy)) +
  geom_polygon(color = "black") +
  scale_fill_viridis_c(option = "magma",
                       na.value = "gray90") +
  theme_minimal() +
  labs(title = "Life Expectancy by Country (2015)",
       fill = "Life Expectancy") +
  coord_fixed(1.3)
```





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Data inconsistencies

```r
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                      "American Samoa"
##  [3] "Andorra"                          "Anguilla"
##  [5] "Antarctica"                       "Antigua"
##  [7] "Ascension Island"                 "Azores"
##  [9] "Barbuda"                          "Bermuda"
## [11] "Bonaire"                          "Canary Islands"
## [13] "Cayman Islands"                   "Chagos Archipelago"
## [15] "Christmas Island"                 "Cocos Islands"
## [17] "Cook Islands"                     "Curacao"
## [19] "Democratic Republic of the Congo" "Dominica"
## [21] "Falkland Islands"                 "Faroe Islands"
## [23] "French Guiana"                    "French Southern and Antarctic Lands"
## [25] "Grenadines"                       "Guadeloupe"
## [27] "Guam"                             "Guernsey"
## [29] "Heard Island"                     "Isle of Man"
## [31] "Ivory Coast"                      "Jersey"
## [33] "Kosovo"                           "Kyrgyzstan"
## [35] "Laos"                             "Liechtenstein"
## [37] "Madeira Islands"                  "Marshall Islands"
## [39] "Martinique"                       "Mayotte"
## [41] "Micronesia"                       "Monaco"
## [43] "Montserrat"                       "Myanmar"
## [45] "Nauru"                            "Nevis"
```

```r
levels(gapminder_2015$country)
```

```
##   [1] "Albania"                  "Algeria"
##   [3] "Angola"                   "Antigua and Barbuda"
##   [5] "Argentina"                "Armenia"
##   [7] "Aruba"                    "Australia"
##   [9] "Austria"                  "Azerbaijan"
##  [11] "Bahamas"                  "Bahrain"
##  [13] "Bangladesh"               "Barbados"
##  [15] "Belarus"                  "Belgium"
##  [17] "Belize"                   "Benin"
##  [19] "Bhutan"                   "Bolivia"
##  [21] "Bosnia and Herzegovina"   "Botswana"
##  [23] "Brazil"                   "Brunei"
##  [25] "Bulgaria"                 "Burkina Faso"
##  [27] "Burundi"                  "Cambodia"
##  [29] "Cameroon"                 "Canada"
##  [31] "Cape Verde"               "Central African Republic"
##  [33] "Chad"                     "Chile"
##  [35] "China"                    "Colombia"
##  [37] "Comoros"                  "Congo, Dem. Rep."
##  [39] "Congo, Rep."              "Costa Rica"
##  [41] "Cote d'Ivoire"            "Croatia"
##  [43] "Cuba"                     "Cyprus"
##  [45] "Czech Republic"           "Denmark"
##  [47] "Djibouti"                 "Dominican Republic"
##  [49] "Ecuador"                  "Egypt"
##  [51] "El Salvador"              "Equatorial Guinea"
##  [53] "Eritrea"                  "Estonia"
```

Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                       "American Samoa"
##  [3] "Andorra"                           "Anguilla"
##  [5] "Antarctica"                        "Antigua"
##  [7] "Ascension Island"                  "Azores"
##  [9] "Barbuda"                           "Bermuda"
## [11] "Bonaire"                           "Canary Islands"
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## [21] "Falkland Islands"                  "Faroe Islands"
## [23] "French Guiana"                     "French Southern and Antarctic Lands"
## [25] "Grenadines"                        "Guadeloupe"
## [27] "Guam"                              "Guernsey"
## [29] "Heard Island"                      "Isle of Man"
## [31] "Ivory Coast"                       "Jersey"
## [33] "Kosovo"                            "Kyrgyzstan"
## [35] "Laos"                              "Liechtenstein"
## [37] "Madeira Islands"                   "Marshall Islands"
## [39] "Martinique"                        "Mayotte"
## [41] "Micronesia"                        "Monaco"
## [43] "Montserrat"                        "Myanmar"
## [45] "Nauru"                             "Nevis"
```

```
levels(gapminder_2015$country)
```

```
##   [1] "Albania"                  "Algeria"
##   [3] "Angola"                   "Antigua and Barbuda"
##   [5] "Argentina"                "Armenia"
##   [7] "Aruba"                    "Australia"
##   [9] "Austria"                  "Azerbaijan"
##  [11] "Bahamas"                  "Bahrain"
##  [13] "Bangladesh"               "Barbados"
##  [15] "Belarus"                  "Belgium"
##  [17] "Belize"                   "Benin"
##  [19] "Bhutan"                   "Bolivia"
##  [21] "Bosnia and Herzegovina"   "Botswana"
##  [23] "Brazil"                   "Brunei"
##  [25] "Bulgaria"                 "Burkina Faso"
##  [27] "Burundi"                  "Cambodia"
##  [29] "Cameroon"                 "Canada"
##  [31] "Cape Verde"               "Central African Republic"
##  [33] "Chad"                     "Chile"
##  [35] "China"                    "Colombia"
##  [37] "Comoros"                  "Congo, Dem. Rep."
##  [39] "Congo, Rep."              "Costa Rica"
##  [41] "Cote d'Ivoire"            "Croatia"
##  [43] "Cuba"                     "Cyprus"
##  [45] "Czech Republic"           "Denmark"
##  [47] "Djibouti"                 "Dominican Republic"
##  [49] "Ecuador"                  "Egypt"
##  [51] "El Salvador"              "Equatorial Guinea"
##  [53] "Eritrea"                  "Estonia"
```

Data inconsistencies

```
issues <- anti_join(world_map, gapminder_2015, by = c("region" = "country"))
levels(as.factor(issues$region))
```

```
##  [1] "Afghanistan"                      "American Samoa"
##  [3] "Andorra"                          "Anguilla"
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## [21] "Falkland Islands"                 "Faroe Islands"
## [23] "French Guiana"                    "French Southern and Antarctic Lands"
## [25] "Grenadines"                       "Guadeloupe"
## [27] "Guam"                             "Guernsey"
## [29] "Heard Island"                     "Isle of Man"
## [31] "Ivory Coast"                      "Jersey"
## [33] "Kosovo"                           "Kyrgyzstan"
## [35] "Laos"                             "Liechtenstein"
## [37] "Madeira Islands"                  "Marshall Islands"
## [39] "Martinique"                       "Mayotte"
## [41] "Micronesia"                       "Monaco"
## [43] "Montserrat"                       "Myanmar"
## [45] "Nauru"                            "Nevis"
```

```
levels(gapminder_2015$country)
```

```
##   [1] "Albania"                  "Algeria"
##   [3] "Angola"                   "Antigua and Barbuda"
##   [5] "Argentina"                "Armenia"
##   [7] "Aruba"                    "Australia"
##   [9] "Austria"                  "Azerbaijan"
##  [11] "Bahamas"                  "Bahrain"
##  [13] "Bangladesh"               "Barbados"
##  [15] "Belarus"                  "Belgium"
##  [17] "Belize"                   "Benin"
##  [19] "Bhutan"                   "Bolivia"
##  [21] "Bosnia and Herzegovina"   "Botswana"
##  [23] "Brazil"                   "Brunei"
##  [25] "Bulgaria"                 "Burkina Faso"
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##  [39] "Congo, Rep."              "Costa Rica"
##  [41] "Cote d'Ivoire"            "Croatia"
##  [43] "Cuba"                     "Cyprus"
##  [45] "Czech Republic"           "Denmark"
##  [47] "Djibouti"                 "Dominican Republic"
##  [49] "Ecuador"                  "Egypt"
##  [51] "El Salvador"              "Equatorial Guinea"
##  [53] "Eritrea"                  "Estonia"
```


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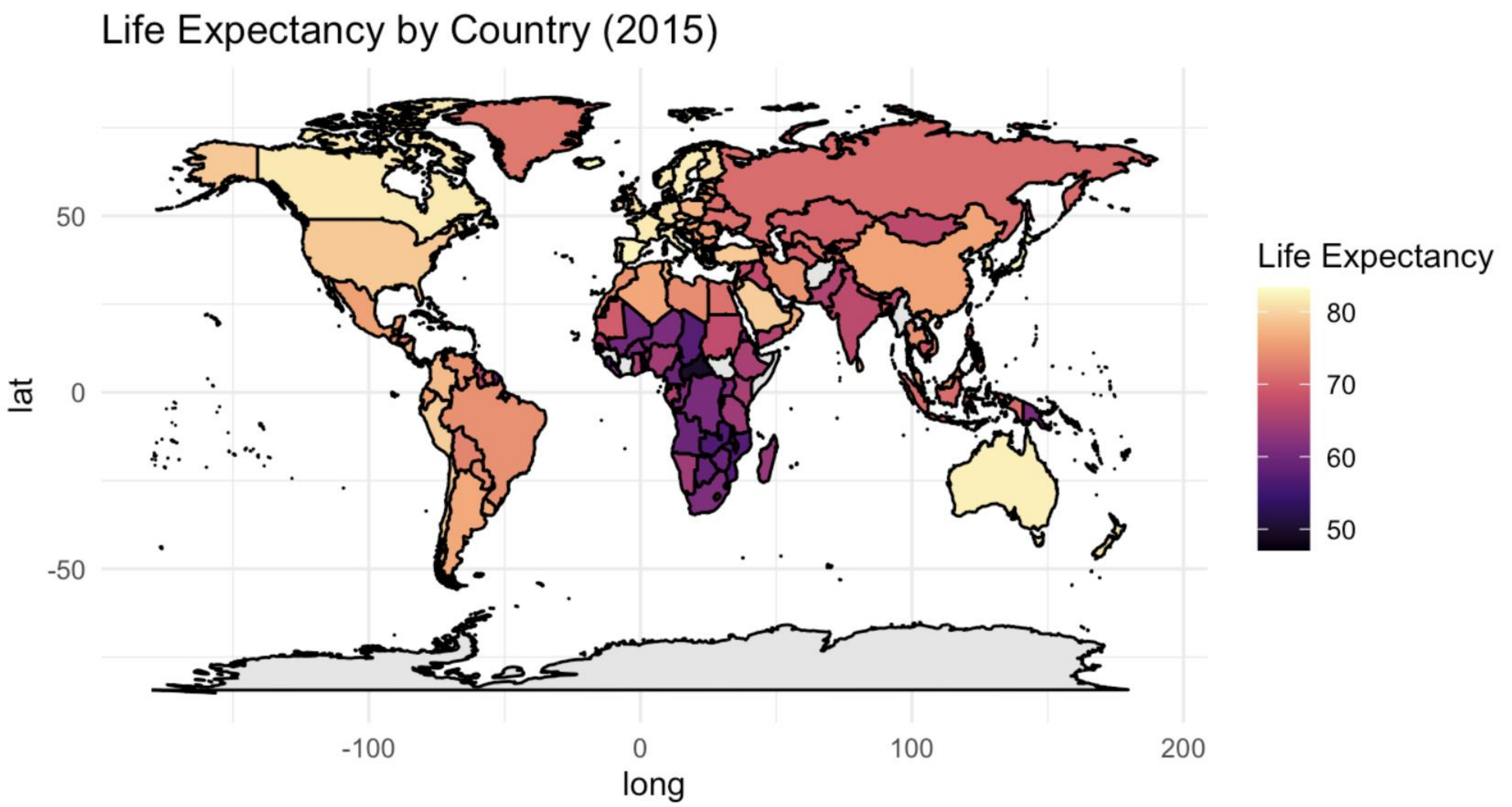
Renaming

```r
gapminder_2015 <- gapminder %>% filter(year == 2015) %>%
  mutate(country =
           revalue(country, c("Antigua and Barbuda" = "Antigua",
                              "Samoa" = "American Samoa",
                              "Congo, Rep." = "Republic of Congo",
                              "Congo, Dem. Rep." = "Democratic Republic of the Congo",
                              "Guinea" = "French Guiana",
                              "Kyrgyz Republic" = "Kyrgyzstan",
                              "St. Lucia" = "Saint Lucia",
                              "St. Vincent and the Grenadines" = "Saint Vincent",
                              "Trinidad and Tobago" = "Trinidad",
                              "United Kingdom" = "UK",
                              "United States" = "USA")))

world_map <- map_data("world")
merged_world <- left_join(world_map, gapminder_2015, by = c("region" = "country"))
```

```r
ggplot(merged_world, aes(long, lat, group = group, fill = life_expectancy)) +
  geom_polygon(color = "black") +
  scale_fill_viridis_c(option = "magma", na.value = "gray90") +
  theme_minimal() +
  labs(title = "Life Expectancy by Country (2015)", fill = "Life Expectancy") +
  coord_fixed(1.3)
```



Summary

- Inconsistencies in the spelling of locations can lead to missing data in plots of maps
- Renaming the locations in one data frame before joining to a second data frame can solve a lot of inconsistencies
- Sometimes new data will need to be collected to complete a map plot